

Comprehensive Financial Architecture, Systems Audit & Implementation Blueprint

Procurement (Purchases), Intangibles & Brand Value, Partner Equity & Drawings, Revenue Treasury, Client A/R & Advance Billing

Target System: DBS-System (FastAPI / Next.js / PostgreSQL)	Audit Date: September 2026
Scope: Financial Integrity, POS, Procurement, Equity & Compliance	Classification: Confidential Executive Advisory

1. Executive Summary & Critical Risk Assessment

A rigorous architectural and codebase audit of the **Aura Luxury Clinic** application revealed that while the system features functional UI screens for checkout and expense recording, **it lacks the structural mechanisms of an audit-ready medical ERP**. The system behaves as an ephemeral cash tracker with four catastrophic failure points: historical salary expense destruction, a primary key collision halting recurring payroll, an authorization deadlock blocking vendor deletions, and permanent de-synchronization between transaction edits and client records. Furthermore, revenue is not stored in any treasury accounts, partner equity is completely missing, advance package billing is unaccounted for, and vendor returns lack debit note mechanisms.

Severity	Identified Vulnerability	Component File	Business & Legal Impact
CRITICAL (P0)	Historical Paid Salary Purge Deleting or setting staff inactive deletes ALL historical paid salaries.	salary.py:L54-64 staff.py:L124, 141	Permanent loss of prior-year tax and operational expense records; invalidates fiscal tax returns.
CRITICAL (P0)	Month 2+ Salary Sync DB Crash Hardcoded ID EXP-SAL-{id} causes PK collision.	salary.py:L39-50	Database crashes on recurring payroll runs as soon as Month 1 salary is paid.
CRITICAL (P0)	Vendor Deletion Deadlock Endpoint requires Admin role, but code demands Partner approval.	expenses.py:L90-120 deps.py:L108	Partners receive 403 Forbidden; 100% approval threshold can never be reached.
HIGH (P1)	Ghost Appointments (Revenue Leakage) Appointments have no payment check or invoice link.	appointment.py:L1-24	Patients receive treatments marked 'Completed' without front-desk POS billing.
HIGH (P1)	Transaction Mutation De-sync Editing invoice amounts does not update Client total spent.	transactions.py:L52-71	Client CRM spend permanently diverges from actual ledger totals.
MEDIUM (P2)	Client-Side Memory Lag Entire transaction/expense history computed in React reduce().	finance-reports/page.tsx:L369-447	Severe browser freezes and network payload bloat as transaction records scale.

2. Deep Dive: Critical Showstopper Bugs

2.1 Catastrophic Historical Salary Expense Purge

Code Trace: In `api/app/services/salary.py` (L54-64), the cleanup function `remove_expenses_by_staff_id()` is executed whenever a staff member's status is toggled to `Inactive` or when they are deleted in `api/app/routers/staff.py`. The query explicitly performs: `select(ExpenseItem).where(ExpenseItem.staff_id == staff_id, ExpenseItem.category == 'Salary')` and deletes **all matching records indiscriminately**, without checking if `status == 'Pending'` or `status == 'Paid'`.

Impact: If a senior doctor worked at the clinic for 24 months and drew Rs. 200,000/month, setting their status to `Inactive` permanently purges Rs. 4,800,000 of historical paid expenses from the database. Prior fiscal year audits and tax filings become instantly falsified.

2.2 Duplicate Primary Key Collision on Month 2+ Payroll

Code Trace: In `salary.py` (L40), newly generated salary records receive a deterministic primary key string: `new_expense = ExpenseItem(id=f'EXP-SAL-{member.id}', ...)`. In Month 1, this record is created and eventually marked `'Paid'`. When Month 2 arrives, `sync_staff_salary_expense()` searches for a *pending* salary, finds none, and attempts to insert Month 2's salary using the identical key `EXP-SAL-{member.id}`. PostgreSQL immediately aborts the transaction with: `psycopg2.errors.UniqueViolation: duplicate key value violates unique constraint 'expenses_pkey'`.

2.3 Authorization Deadlock in Multi-Approval Vendor Expense Deletions

Code Trace: In `api/app/routers/expenses.py` (L90-120), deleting a vendor expense mandates 100% sign-off from all accounts with roles `admin` and `partner`: `all(email in current_approvals for email in required_emails)`. However, the FastAPI endpoint dependency is defined as: `current_user = Depends(get_admin_user)`. When an equity partner logs in to submit their required approval, the endpoint throws a 403 Forbidden error. Consequently, the approval threshold can never reach 100%, and vendor expenses cannot be deleted.

3. The System of Purchases, Procurement & Accounts Payable

In the current system, purchases of botox vials, hyaluronic fillers, laser consumables, and creams are handled through ad-hoc manual entries in the general expenses table. Stock inventory is manually incremented with no link to purchase costs.

The Required 3-Way Match Procurement Cycle

To prevent medical supply shrinkage, over-billing, and unverified deliveries, all purchases must follow the 3-Way Match workflow:

Step	Workflow Stage	Operational Action	Ledger & Inventory Impact
1	Purchase Order (PO)	Doctor or Inventory Head drafts PO with approved vendor and negotiated unit costs.	No ledger entry. Status: 'Sent to Vendor'. Commitments tracked.
2	Goods Received Note (GRN)	Stock arrives at clinic. Staff inspects boxes, logs physical count, Batch Number and Expiry Date .	Inventory Asset incremented immediately. Consumables available for treatment.
3	Supplier Bill & 3-Way Match	Vendor sends invoice. System verifies PO Price == Bill Price and Ordered Qty == GRN Received Qty.	Debit: Inventory Asset (#1300) Credit: Accounts Payable to Supplier (#2010)
4	Bill Payment / Settlement	Clinic issues bank transfer or cheque according to supplier credit terms (e.g. Net 30).	Debit: Accounts Payable (#2010) Credit: Operating Bank Account (#1030)

4. Retail Product Sales & Vendor Purchase Returns (Debit Notes)

Luxury aesthetic clinics operate two distinct inventory streams: **Internal Professional Consumables** (vials and needles used in procedures) and **Retail Skincare Merchandise** (serums and sunscreens sold to take home). Furthermore, surplus, near-expiry, or damaged goods must be returned to suppliers with enforceable financial recovery.

4.1 Retail Product Sales Engine

- **Dual-Catalog Separation:** Retail products must feature Barcode/SKU scanning at POS, independent retail prices, and wholesale purchase costs.
- **Automated COGS & Gross Margin:** Selling a retail cream for Rs. 8,000 that was purchased at wholesale for Rs. 4,500 automatically generates: Debit: Cost of Goods Sold (#5020) Rs. 4,500 and Credit: Inventory Asset (#1300) Rs. 4,500, locking in Rs. 3,500 Gross Margin.
- **Dual-Tax Invoicing:** Medical procedures fall under Provincial Sales Tax (PRA/SRB), whereas retail cosmetic sales fall under Federal Sales Tax (FBR 18%). The POS invoice must itemize and separate these tax codes automatically.

4.2 Vendor Purchase Returns (Return to Vendor - RTV) & Debit Notes

When stock is damaged on delivery, ordered in excess, or nearing expiry, the clinic returns items to the distributor. The system must execute a 4-step Return to Vendor flow:

Action	System Operation	Financial & Accounting Entry
1. Create RTV Order	Manager selects vendor, items, quantity returned, and return reason (e.g. 'Near Expiry').	Immediately decrements physical units from inventory ledger.
2. Generate Debit Note	System prints official Debit Note (DN-2026-001) with serial number for supplier courier sign-off.	Serves as legal proof of returned pharmaceutical goods.
3. A/P Settlement (Choice A)	If the vendor invoice was unpaid (Credit purchase), deduct returned value from outstanding bill.	Debit: Accounts Payable (#2010) Credit: Inventory Asset (#1300)
4. Refund / Credit (Choice B)	If bill was already paid, record incoming cash/bank refund or hold as Vendor Credit Balance.	Debit: Clinic Bank Account (#1030) Credit: Inventory Asset (#1300)

5. Advance Payments for Packages & Client Accounts Receivable (Dues)

Aesthetic clinics thrive on multi-session treatment packages and frequently extend credit terms to VIP clients. The platform currently has zero support for either flow.

5.1 Multi-Session Packages & Deferred Revenue (GAAP / IFRS 15)

When a client pays Rs. 45,000 advance for a 3-session series (Rs. 15,000/session):

- **Day 1 (Advance Paid):** Money received is **NOT** immediate earned revenue. It must be credited to a liability account: Debit: Bank (#1030) Rs. 45,000 / Credit: Deferred Revenue Liability (#2050) Rs. 45,000. A ClientPackage record is initialized with 3 Total Sessions and 0 Used.
- **Day 30 (Session 1 Redeemed):** Client arrives. Front desk selects 'Redeem Session'. The system records: Debit: Deferred Revenue (#2050) Rs. 15,000 / Credit: Treatment Revenue (#4010) Rs. 15,000. A zero-charge receipt prints showing: *'Session 1 of 3 Redeemed — 2 Sessions Remaining'*.

5.2 Client Accounts Receivable (A/R) & Installment Dues Ledger

When a client receives treatments worth Rs. 50,000, pays Rs. 20,000 cash today, and owes Rs. 30,000 on next visit:

- **Split Invoice at Checkout:** The transaction logs: Amount Paid Today: Rs. 20,000, Remaining Due: Rs. 30,000, and Status: Partially_Paid. Ledger: Debit: Cash Rs. 20,000, Debit: Client Accounts Receivable (#1200) Rs. 30,000, Credit: Service Revenue (#4010) Rs. 50,000.
- **Client Profile Statement & Settlement:** In the Client profile, an 'Outstanding Dues' badge displays Rs. 30,000. When the client returns to settle, clicking 'Collect Due Payment' issues an official Payment Receipt Voucher, clearing the debt: Debit: Bank/Cash (+Rs. 30,000) / Credit: Accounts Receivable (-Rs. 30,000).

6. Where Monthly Revenue Gets Stored: Treasury, Vaults & Month-End Close

In the existing codebase, revenue is **phantom money**: it exists only as an on-the-fly SQL summation over flat rows. It does not sit in any real-world asset account, and at month's end, it is never closed into retained earnings.

6.1 Multi-Tier Treasury Architecture

Every rupee collected by the POS must map to a verified physical or digital treasury asset account:

Treasury Tier	Account Type & GL Code	Physical / Digital Location	Reconciliation Cycle
Tier 1: Drawer	Front Desk Cash Drawer (#1010 - Asset)	Physical cash register at branch reception desk.	Daily Z-Report: Cashier counts physical cash at shift end. Over/short variances logged.
Tier 2: Vault	Clinic Main Safe / Vault (#1015 - Asset)	Dual-custody physical safe in manager's office.	Weekly Sweep: Excess branch drawer cash transferred to vault for security.
Tier 3: In-Transit	POS Merchant Settlement (#1020 - Asset)	Credit card & online transaction funds held by bank terminal.	T+1 Settlement: Reconciled automatically against bank terminal batch reports.
Tier 4: Bank	Commercial Operating Acc (#1030 - Asset)	Meezan Bank / HBL Corporate Current Account.	Monthly Bank Reconciliation: Bank statement synced against general ledger.

6.2 The Month-End Closing Engine (Closing to Retained Earnings)

At 11:59 PM on the final day of each calendar month, the platform must execute a formal fiscal period close:

- All temporary revenue accounts (#4010, #4020) are debited (zeroed out).
- All temporary operational expense accounts (#5010-#5090) are credited (zeroed out).
- Net Income is calculated and credited to **Retained Earnings (#3010)** on the Balance Sheet.
- A specified percentage (e.g. 30%) is retained in the clinic for laser maintenance and working capital reserves; the remaining 70% flows into the **Partner Distributable Profit Pool**.

7. Partners' Equity Dashboard, Withdrawals & Brand Valuation

Partners are currently represented merely as plain usernames in the users table. A luxury clinic must provide equity partners complete transparency into two vital metrics: **(1) Exactly how much money each partner has withdrawn to date**, and **(2) Exactly what their equity stake in the clinic brand is worth right now**.

7.1 Tracking Partner Withdrawals (Drawings Account)

- **Separation from Expenses:** Partner withdrawals are **NOT** clinic operational expenses. Logging a partner withdrawal as an expense is illegal under tax law and corrupts clinic P&L; margins.
- **Equity Reduction:** Every drawing event (cheque, personal card payment, cash draw) is debited to the partner's Partner Drawings Account (#3020), directly reducing their personal capital balance.
- **Audit Trail:** Each withdrawal records: Date, Amount, Disbursing Account (e.g. Meezan Bank), Purpose, and Sign-off.

7.2 What Partners Have in the Brand Right Now: Book vs. Brand Value

A partner's financial stake in Aura Luxury Clinic consists of two distinct valuation layers:

Valuation Metric	Calculation Formula	Practical Meaning for the Partner
1. Book Capital Balance (Net Liquidation Value)	Initial Capital Injected + Share of Accumulated Net Profit - Total Cumulative Drawings	The exact cash and accounting equity the partner owns on the clinic balance sheet right now. Can be drawn as dividends or held as capital.

2. Fair Market Brand Value (Enterprise Stake)	(Trailing 12-Mo EBITDA × Multiple [e.g. 5x] + Net Tangible Assets) × Partner Equity %	What the partner's ownership stake in the clinic is worth if sold, franchised, or pitched to external healthcare investors today.
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7.3 Visual Partner Equity Dashboard Layout Specification

In app/finance-reports/page.tsx, a dedicated 'Partner Equity & Brand Valuation' tab will render the following card grid:

Metric	Partner A (Managing Partner — 60%)	Partner B (Investing Partner — 40%)	Consolidated Clinic Brand
Initial Capital Injected	Rs. 6,000,000	Rs. 4,000,000	Rs. 10,000,000
Cumulative Profit Share Earned	Rs. 7,200,000 (60% of Net)	Rs. 4,800,000 (40% of Net)	Rs. 12,000,000
Total Withdrawn to Date (Drawings)	-Rs. 4,500,000	-Rs. 2,000,000	-Rs. 6,500,000
Current Net Capital Balance (Book)	Rs. 8,700,000	Rs. 6,800,000	Rs. 15,500,000
Current Market Brand Stake (5x EBITDA)	Rs. 24,000,000	Rs. 16,000,000	Rs. 40,000,000

8. Technical Remediation & Implementation Roadmap

To transform DBS-System into an enterprise luxury clinic platform, development should follow this phased implementation timeline:

Phase	Focus Area	Key Technical Deliverables	Estimated Effort
Phase 1 (Hotfixes)	Critical Bug Patches & Data Integrity	- In salary.py: modify remove_expenses_by_staff_id() to only purge status == 'Pending'. - Generate date-tokenized salary IDs (EXP-SAL-{id}-{YYYYMM}) to eliminate duplicate PK collisions. - Change DELETE /expenses/{id} dependency to get_admin_or_partner_user. - Sync Client.total_spent on PUT /transactions/{id}.	48 Hours (Immediate P0)
Phase 2 (POS & Billing)	Packages, Dues & Retail Products	- Implement ClientPackage and deferred session redemption tracking. - Add amount_paid and remaining_due to transactions for partial client billing. - Add Client Profile Outstanding Dues statement and Installment Receipt generation. - Decouple retail take-home products with barcode lookup and automated wholesale COGS.	Weeks 1 – 2
Phase 3 (Purchases & RTV)	Procurement & Vendor Returns	- Build Purchase Order (PO) and Goods Received Note (GRN) workflow with batch & expiry dates. - Implement Return to Vendor (RTV) interface with printable Debit Notes. - Real-time Accounts Payable (A/P) aging statements for pharmaceutical suppliers.	Weeks 3 – 4
Phase 4 (Treasury & Equity)	Treasury, Close & Partner Dashboard	- Setup multi-account Treasury (Cash Drawers, Main Vault, Commercial Bank Accounts). - Build End-of-Day Cash Drawer Z-Report shift closeout screen. - Implement Automated Month-End Close allocating net profit into Retained Earnings. - Deploy Partner Equity Dashboard displaying capital, withdrawals to date, and live brand valuation.	Weeks 5 – 6

Sign-Off & Document Verification

This document represents the complete, unified financial and technical audit for Aura Luxury Clinic. Executing the Phase 1 hotfixes immediately guarantees that no historical salary records will be purged, and paves the way for a fully compliant, multi-branch, partner-governed luxury medical ERP.